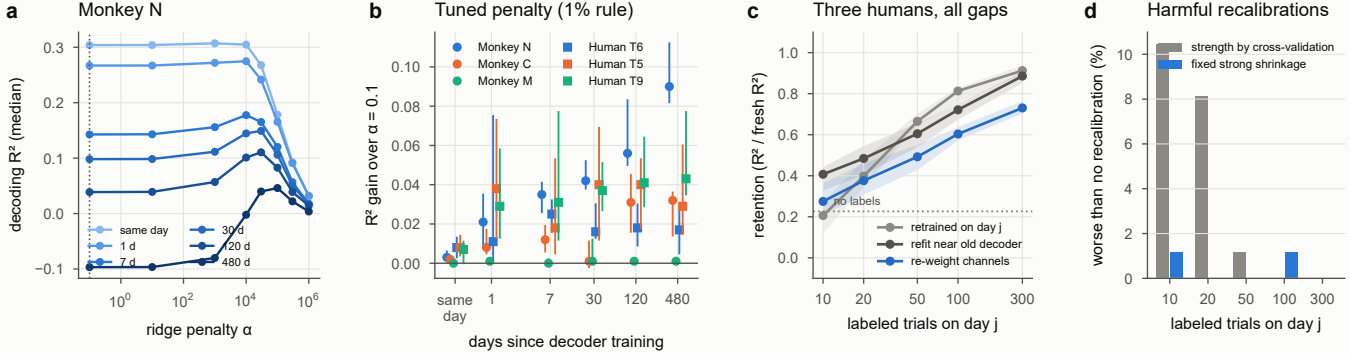

a
Monkey N
decoding R² (median)
0.3
0.2
0.1
0.0
−0.1
same day
1 d
7 d
30 d
120 d
480 d
10^0
10^2
10^4
10^6
ridge penalty α
b
Tuned penalty (1% rule)
Monkey N
Monkey C
Monkey M
Human T6
Human T5
Human T9
R² gain over α = 0.1
0.10
0.08
0.06
0.04
0.02
0.00
same day
1
7
30
120
480
days since decoder training
c
Three humans, all gaps
retention (R² / fresh R²)
1.0
0.8
0.6
0.4
0.2
0.0
no labels
retrained on day j
refit near old decoder
re-weight channels
10
20
50
100
300
labeled trials on day j
d
Harmful recalibrations
worse than no recalibration (%)
10
8
6
4
2
0
strength by cross-validation
fixed strong shrinkage
10
20
50
100
300
labeled trials on day j